Modular Structural Arithmetic (MSA): Clean Formulation

0. Notation and Basic Objects

- Let $\mathbb Z$ denote the integers. For a prime p, write $\mathbb Z/p\mathbb Z$ for the residue ring and $(\mathbb Z/p\mathbb Z)^\times$ for its multiplicative group.
- Reduction mod p is $\rho_p:\mathbb Z\to\mathbb Z/p\mathbb Z$, $\rho_p(x)\equiv x\pmod p$.
- The doubling map is $\delta:\mathbb Z\to\mathbb Z$, $\delta(x)=2x$.
- Distinguished constants: $\gamma=24$, $\mu=48$, $\varepsilon=96$.

1. Induced Modular Dynamics

For each prime p, define $\delta_p:\mathbb Z/p\mathbb Z\to\mathbb Z/p\mathbb Z$ by $\delta_p([x])=[2x]$.

Theorem 1 (Doubling commutes with reduction)

For every prime p and every $x\in\mathbb Z$,

$$\rho_p(\delta(x))=\delta_p(\rho_p(x)).$$

Proof. $\rho_p(2x)=[2][x]=\delta_p([x])$ by ring homomorphism properties. $\square$

We abbreviate this global commutation property as **DP(p)**.

Definition 1 (Order of doubling mod p)

For an odd prime p, let $r=\mathrm{ord}_p(2)$ be the least positive integer with $2^r\equiv1\pmod p$.

Theorem 2 (Resonance length)

For any odd prime p with $r=\mathrm{ord}_p(2)$ and any $x\in\mathbb Z$,

$$\delta^r(x)\equiv x\pmod p.$$

In particular, every orbit of δ_p in $\mathbb Z/p\mathbb Z$ has period dividing r; orbits inside $(\mathbb Z/p\mathbb Z)^\times$ are cycles whose lengths divide r.

Proof. $\delta^r(x)=2^r x\equiv x\pmod p$. $\square$

Lemma 1 (Distinctness of seeds)

If $p\ge5$ is prime, then $[\gamma],[\mu],[\varepsilon]$ are nonzero and pairwise distinct in $\mathbb Z/p\mathbb Z$.

Proof. $p\nmid 24,48,96$ and $\mu-\gamma=24$, $\varepsilon-\mu=48$ are not divisible by any prime ≥5. $\square$

2. Role System and Modularity

Introduce three unary predicates on nonzero integers

$$\mathrm{Gen}(x),\quad \mathrm{Med}(x),\quad \mathrm{Man}(x),\qquad x\in\mathbb Z\setminus\{0\}.$$

Axiom R1 (Partition)

For every $x\ne0$, exactly one of $\mathrm{Gen}(x)$, $\mathrm{Med}(x)$, $\mathrm{Man}(x)$ holds.

Axiom R2 (Seed labels)

$\mathrm{Gen}(\gamma)$, $\mathrm{Med}(\mu)$, and $\mathrm{Man}(\varepsilon)$ hold.

Axiom R3 (Congruence compatibility on units)

For every odd prime p, if $x\equiv y\pmod p$ and $xy\not\equiv0\pmod p$, then

$$\mathrm{Gen}(x)\Leftrightarrow\mathrm{Gen}(y),\quad \mathrm{Med}(x)\Leftrightarrow\mathrm{Med}(y),\quad \mathrm{Man}(x)\Leftrightarrow\mathrm{Man}(y).$$

Equivalently, the role map factors through ρ_p on $\mathbb Z\setminus p\mathbb Z$.

Definition 2 (Structural property SP(p))

For a prime p, **SP(p)** holds if: 1. p is odd; 2. $[\gamma],[\mu],[\varepsilon]$ are nonzero and pairwise distinct in $\mathbb Z/p\mathbb Z$ (automatic for $p\ge5$ by Lemma 1); 3. Axiom R3 applies at p.

Under SP(p), roles induce well-defined predicates on $(\mathbb Z/p\mathbb Z)^\times$:

$$\mathrm{Gen}_p([x])\iff\mathrm{Gen}(x),\quad \mathrm{Med}_p([x])\iff\mathrm{Med}(x),\quad \mathrm{Man}_p([x])\iff\mathrm{Man}(x).$$

Theorem 3 (Role partition mod p)

If SP(p) holds, then ${\mathrm{Gen}_p,\mathrm{Med}_p,\mathrm{Man}_p}$ is a disjoint partition of $(\mathbb Z/p\mathbb Z)^\times$.

Proof. By R1 on representatives and R3 for well-definedness. $\square$

3. Dynamics and Roles

No role-dynamical constraints are assumed by default. When needed, one may adopt the following optional compatibility, which then propagates labels along doubling orbits.

Optional Axiom D1 (Doubling invariance of roles)

For all $x\ne0$, $\mathrm{Gen}(x)\Leftrightarrow\mathrm{Gen}(2x)$, and similarly for Med and Man.

Under D1 and SP(p), roles are constant on δ_p-orbits in $(\mathbb Z/p\mathbb Z)^\times$.

4. Consolidated Statements

- **DP(p)** holds for every prime p (Theorem 1).
- For every odd prime p, the resonance length $r=\mathrm{ord}_p(2)$ controls all δ_p-orbit periods (Theorem 2).
- For every prime $p\ge5$, seeds $[\gamma],[\mu],[\varepsilon]$ are distinct and nonzero (Lemma 1). Hence SP(p) holds once R3 is adopted.
- If SP(p) holds, then roles descend to $(\mathbb Z/p\mathbb Z)^\times$ and form a partition (Theorem 3).
- If D1 is also assumed, each δ_p-orbit sits entirely inside one of the three role classes.

5. Minimal Data to Instantiate MSA

1. Base set and operator: $(\mathbb Z,\delta)$ with $\delta(x)=2x)$.
2. Distinguished constants: $\gamma=24,\ \mu=48,\ \varepsilon=96$.
3. Role predicates $\mathrm{Gen},\mathrm{Med},\mathrm{Man}$ satisfying R1–R2, optionally D1.
4. For modular analysis at prime p: invoke DP(p) always; invoke SP(p) to transport roles to $(\mathbb Z/p\mathbb Z)^\times$; compute $r=\mathrm{ord}_p(2)$ to describe δ_p-dynamics.

Remark. The framework separates arithmetic facts (DP(p), resonance via $\mathrm{ord}_p(2)$) from structural labeling (R1–R3, optional D1). This allows safe reuse of the arithmetic layer independent of any chosen role semantics.